

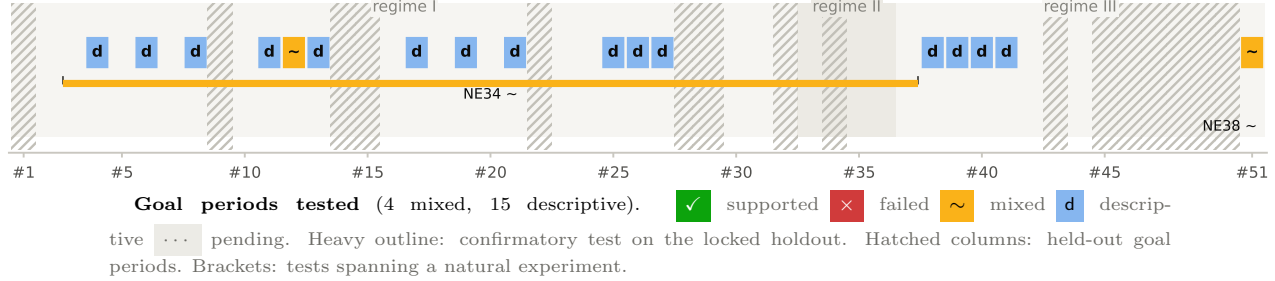
# H105: Goals act on a two-state order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1/5

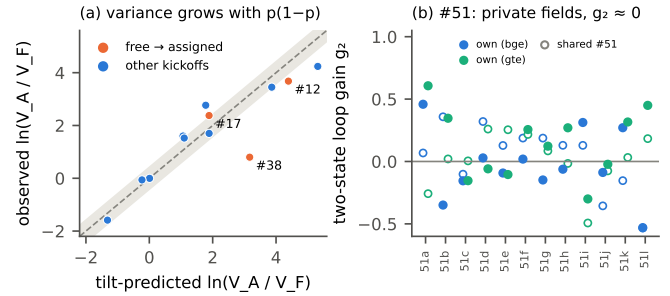
**Bottom line.** Goals act on a two-state occupancy whose variance grows as  $p(1-p)$ , but the free-week tilt forecast is not identifiable at village sampling, so the Legendre question stays open.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

H10's Legendre test failed on continuous alignment. HH130 proposed a fairer test: treat each agent-window as on-goal or off-goal and ask whether a uniform field (tilt) on the free week predicts the assigned week's occupancy variance and who moves. I ran it on three free → assigned pairs, 16 kickoffs and three natives, in two embedding models. The synthetic came first and showed the tilt test is not identifiable at village sampling: with a true tilt it passes in 30% of pairs, and in 40% under a coupling change. The two-state picture holds descriptively. Occupancy rises from the decoy floor (0.02–0.05) to 0.23–0.44, its variance grows as  $p(1-p)$  (Spearman 0.86 over 16 kickoffs), and transverse occupancy stays flat. Who moves: inconclusive in 3/3 pairs. Holdout: not run.



**Observables.** (a) Observed against tilt-predicted growth of the occupancy variance, free (or previous) week to assigned week, for 16 kickoffs; the band is  $\pm \ln 1.5$ . The ranks agree ( $\rho_s = 0.86$ ), but only 7/16 fall in the band; #38a sits far below. (b) #51: loop gain of each agent's spin along its own private goal (filled) and along the shared #51 direction (open), per unit. Private fields do not synchronize, except the role-onset unit 51a.

## PHYSICAL MODEL

Model 01, forward mean field (Curie–Weiss with agent fields), in 0/1 spins. Spin  $\sigma_{i,t}$ : agent  $i$ 's statements in 30-min window  $t$  are mostly on-goal (projection on  $\hat{g}$  above the 95th percentile of decoy statements). Field  $h_i$ : each agent's on-goal tendency. Coupling  $J_2$ : synchronized switching. Occupancy  $p_i$ : mean spin. The tilt adds one  $\lambda$  to every  $h_i$  and keeps  $J_2$ .  $g_2$ : two-state loop gain.

$$P(\sigma) \propto \exp\left(\sum_i h_i \sigma_i + \frac{J_2}{2N} n^2\right), \quad n = \sum_i \sigma_i$$

$$\text{Var}(n) = \frac{S}{1 - J_2 \bar{q}}, \quad S = \sum_i p_i(1 - p_i), \quad g_2 = J_2 \bar{q}$$

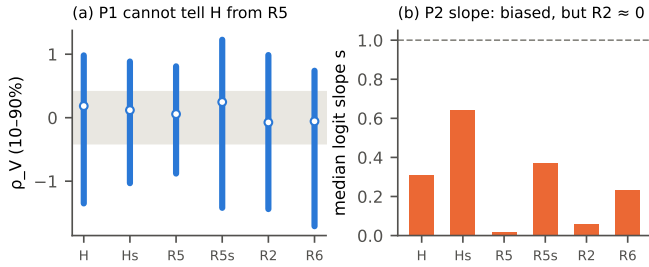
$$\text{tilt: } h_i \rightarrow h_i + \lambda \Rightarrow \rho_V = \ln \frac{V_A^{\text{obs}}}{V_A(\lambda, J_2^F)} \approx 0$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a methods result. The two-state view is the right bookkeeping for a goal: it moves only the on-goal occupancy, from about 0.03 to 0.2–0.45, and leaves other directions alone. But the forecast HH130 wanted, steerability from a free week, cannot be made from village logs with this instrument: free weeks have almost no on-goal windows, so their coupling is unmeasurable, and 2–4 statements per window misclassify the spin. An operator learns nothing new beyond H10 and H54. A monitor builder learns that occupancy measured this way needs a calibrated classifier before any fluctuation–response forecast.

## RESULTS VS PREDICTIONS

| Prediction             | Observed                                      | Verdict          |
|------------------------|-----------------------------------------------|------------------|
| P1 tilt variance       | $\rho_V -0.71, +0.53, -2.45$ ; gte flips #12a | not identifiable |
| P2 uniform logit shift | $s -0.60, +0.38, +2.53$ ; calibrated 0/3      | inconclusive     |
| P3 $J_2$ unchanged     | $\Delta g_2 +0.64, +0.26, +0.03$              | not identifiable |
| P4 transverse flat     | $p_\perp \approx 0.02$ in F and A             | supported        |
| P6 kickoffs            | 7/16 in band; $\rho_s 0.86$ ( $p 0.0003$ )    | half / half      |
| N1 G51 private goals   | $g_2 -0.08$ (bge), $+0.19$ (gte)              | mixed            |
| N2 G12 debate schedule | removes 50% of excess (gte); no excess (bge)  | mixed            |
| N3 NE38 one spin       | $0 \rightarrow 0.96$ ; others $+0.01 / +0.05$ | mixed            |



**Second figure.** Latent Curie-Weiss spins at the real window structure of the three pairs, observed through statements as in the real pipeline (30 replicates per scenario). (a) The tilt (H) and a coupling change (R5) give the same  $\rho_V$  spread. (b) The logit slope is biased low under the tilt (0.31) but near 0 under a common target (R2), so P2 keeps some resolution.

## GOAL PERIODS IN DETAIL

G04 d descriptive  
G06 d descriptive  
G08 d descriptive  
G11 d descriptive  
G12 ~ mixed  
G13 d descriptive  
G17 d descriptive  
G19 d descriptive  
G21 d descriptive  
G25 d descriptive  
+9 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 2 |
| B assumptions          | 0 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 0 |
| D unfitted predictions | 0 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

"On-goal" is a text-embedding threshold; the 90% and 98% variants shift  $p$  but not the conclusions. Spins built from 2-4 statements are often wrong, which drives the non-identifiability. The two models disagree on the sign of  $\rho_V$  in #12a and on both natives' second clauses. Free weeks are not field-free. The variance-growth agreement across kickoffs follows from any binary occupancy and does not test the tilt. The on-goal work variant was not run.

## ROUND 2

Build a calibrated on-goal classifier with a measured confusion matrix, prove identifiability on the synthetic, and pool free-week couplings hierarchically before retesting the tilt.

- **H105-R1** Fix the instrument, not the model.
- **H105-R2** Pool the free-week coupling hierarchically (exception (d)) across the free weeks of one regime to measure  $J_2$  at all, then predict each assigned week.
- **H105-R3** Separate drive from coupling in #12 with the DQ6 debate schedule as an explicit time-dependent field and a read-out (ledger) lag test for the residual.

H105-two-state-goal-order · generated 2026-10-07 17:49 by infra/summaries/build\_summaries.py · card hypotheses/H105-two-state-goal-order/README.md · ratings: H105 agent (suggested) (page 2)